LOGESH S

Robotics Software Engineer — ROS 2 | Manipulation | Autonomous Navigation

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ROS 2 engineer with hands-on experience in manipulation pipelines (MoveIt 2 / MTC), autonomous navigation (Nav2 / AMCL), and NVIDIA Isaac Sim digital twin development. IEEE-published researcher with experience in building sensor-fused AMR systems from simulation to real hardware deployment.

TECHNICAL SKILLS

Core Robotics: ROS 2, Nav2, MoveIt 2, MTC, BehaviorTree.CPP
State Estimation: EKF, UKF, Particle Filter, AMCL
Simulation: NVIDIA Isaac Sim, Gazebo, Genesis
Programming: C++, Python, Flutter/Dart
DevOps: Docker, Git, AWS
Hardware: Jetson Nano, Raspberry Pi 5
CAD & Design: Fusion 360, SolidWorks, Blender, Adobe Substance Suite

EXPERIENCE

Robotics Engineering Intern — Flo Mobility Pvt. Ltd. Jun 2024 – Jul 2024
HSR Layout, Bengaluru, Karnataka

- Created industry-grade **URDF** models for 2 variants of the Material Moving Robot (MMR) platform, optimizing meshes up to **80%** using convex hull decomposition.
- Deployed and tuned the **ROS 2** navigation stack (**Nav2**) in **Gazebo** across a custom sand dune terrain simulation, optimizing costmaps and localization to achieve **90%+** reliable autonomous navigation in unstructured environments.

PROJECTS

Smart In-House Logistics Via ROS Environment (NIKIRO) Sep 2024 – May 2025
SREC Final Year Project · IEEE ICCRTEE 2025 Published · Hackster.io Featured · github.com/logesh1516/Nikiro

- Designed and deployed a full-stack AMR integrating MyCobot 280 on **Jetson Nano** with **ROS 2**, **Nav2**, and **SLAM Toolbox** for end-to-end intra-facility logistics automation—awarded **Best Final Year Project 2025** and featured on **ROS Discourse**.
- Developed a **Flutter** mobile application integrated with **ROS 2**, enabling real-time telemetry streaming (pose, velocity, battery status), remote robot control, and live occupancy map visualization during autonomous navigation.

Wine Glass Manipulation using MoveIt Task Constructor (MTC) Mar 2026
Independent R&D · github.com/logesh1516/Wine-glass-Manipulation

- Engineered a full pick-and-place manipulation pipeline in **MoveIt 2** + MTC, implementing constraint-aware task stages—grasp pose generation, approach, pick, lift, place, and retreat.
- Validated the pipeline in **NVIDIA Isaac Sim** with mesh-based collision geometry, achieving stable grasp execution on delicate wine glass geometry.

Digital Twin for MyCobot 280 JN Jan 2024 – May 2025
SREC

- Built a high-fidelity digital twin of MyCobot 280 in **NVIDIA Isaac Sim** with **ROS 2** integration for synchronized simulation and hardware-in-the-loop (HIL) validation.
- Developed photorealistic robot models and environments using **Fusion 360**, **Blender**, and Adobe Substance Suite for use across planning and perception testing workflows.

URDF Mesh Optimization for Industrial Manipulators Nov 2025

Independent R&D · github.com/logesh1516/urdf-mesh-optimization

- Documented an end-to-end CAD-to-URDF optimization workflow for 4 industrial manipulators (**KUKA IIWA**, **ABB IRB-5720**, **Yaskawa AR1440**, HC10DTP) covering mesh decimation, normal repair, and visual/collision separation in **Blender**, achieving **80–95%** polygon reduction.
- Produced simulation-ready **ROS 2** description packages with separated visual/collision meshes compatible with **MoveIt 2**, **Gazebo**, and **NVIDIA Isaac Sim**.

EDUCATION

B.E. in Robotics and Automation — CGPA: 8.67 / 10 Nov 2021 – May 2025

Sri Ramakrishna Engineering College, Coimbatore

Minor in Internet of Things — CGPA: 8.37 / 10 Nov 2021 – May 2025

Sri Ramakrishna Engineering College, Coimbatore

AWARDS & RECOGNITION

- **IEEE Publication — ICCRTEE 2025:** "Smart In-House Logistics via ROS Environment" — peer-reviewed and published at Kalasalingam Academy of Research and Education.
- **Best Final Year Project 2025** — Dept. of Robotics, SREC. Recognized for innovation in automating intra-facility logistics using ROS 2.
- **Hackster.io Feature** — Nikiro AMR officially featured by Elephant Robotics on Hackster.io and Open Robotics Discourse.
- **VS Code Extension**—MoveIt Task Constructor Snippets published on VS Code Marketplace; C++ templates for MoveIt 2 Task Constructor.
- **Japanese Subject Topper** — Department of English, SREC (2021–22 Semester 2).

LANGUAGES

English, Tamil, Japanese, Mandarin